Fusion
Technical Project Documentation for a Chat Application

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1 Project Overview

Fusion is a client-server based chat application developed as a semester project. It consists of:

- A WPF desktop client (MVVM architecture)
- A REST-based ASP.NET Core backend
- A PostgreSQL database

The system supports:

- User registration and login
- JWT-based authentication
- Direct and group conversations
- Persistent message storage

2 System Architecture

Fusion follows a layered client-server architecture with strict separation of concerns.

2.1 Architectural Overview

- Presentation layer (WPF UI)
- Application layer (ViewModels and Services)
- API layer (ASP.NET Controllers)
- Persistence layer (EF Core + PostgreSQL)

All communication between client and server is performed via REST endpoints using JSON-based DTOs. The client never communicates directly with the database.

2.2 Layered Separation Principle

Each layer is isolated:

- UI logic (Views)
- State and behavior (ViewModels)
- HTTP communication (API Services)
- Business logic (Controllers)
- Data persistence (Store implementations)

This architecture allows replacing implementations without affecting higher layers.

3 Client Architecture (WPF)

3.1 MVVM Pattern

The client implements the Model-View-ViewModel pattern.

- Models represent UI-bound data
- ViewModels manage state and commands
- Views are implemented in XAML

3.2 Property Change Notification

```
public class ContactModel : ObservableObject
{
    private string _username;

    public string Username
    {
        get => _username;
        set
        {
            _username = value;
            OnPropertyChanged();
        }
    }
}
```

The base class implements `INotifyPropertyChanged` to ensure UI updates.

3.3 Command-Based Interaction

```
public class AsyncRelayCommand : ICommand
{
    private readonly Func<Task> _execute;
    private readonly Func<bool>? _canExecute;
    private bool _isExecuting;

    public async void Execute(object? parameter)
    {
        if (!CanExecute(parameter))
            return;

        try
        {
            _isExecuting = true;
            RaiseCanExecuteChanged();
            await _execute();
        }
        finally
```

```
        {
            _isExecuting = false;
            RaiseCanExecuteChanged();
        }
    }
}
```

This prevents concurrent command execution during asynchronous operations.

3.4 Authentication Flow (Client)

```
public async Task<LoginResult> LoginAsync(string email, string password)
{
    var req = new LoginRequest
    {
        Email = email.Trim(),
        Password = password
    };

    var dto = await PostJsonAsync<LoginRequest, AuthResponse>("auth/login", req
        );

    if (!dto.Success || string.IsNullOrWhiteSpace(dto.AccessToken))
        return LoginResult.Fail("Invalid credentials.");

    Session.AccessToken = dto.AccessToken;

    _http.DefaultRequestHeaders.Authorization =
        new AuthenticationHeaderValue("Bearer", Session.AccessToken);

    return LoginResult.Ok();
}
```

The JWT token is stored in memory and attached to all subsequent HTTP requests.

4 Server Architecture

4.1 JWT Claim Extraction

```
private static Guid GetUserIdFromClaims(ClaimsPrincipal user)
{
    var raw =
        user.FindFirstValue(ClaimTypes.NameIdentifier) ??
        user.FindFirstValue("sub");

    return Guid.TryParse(raw, out var id) ? id : Guid.Empty;
}
```

Identity is derived strictly from validated JWT claims.

4.2 Message Endpoint

```
[HttpPost("{conversationId:guid}/messages")]
public async Task<ActionResult<MessageDto>> SendMessage(
    Guid conversationId,
    [FromBody] SendMessageRequest req)
{
    var me = GetUserIdFromClaims(User);
    if (me == Guid.Empty) return Unauthorized();

    var content = (req.Content ?? "").Trim();
    if (string.IsNullOrWhiteSpace(content))
        return BadRequest("Empty message.");

    var msg = await _chatStore.AddMessageAsync(conversationId, me, content);

    return Ok(new MessageDto
    {
        MessageId = msg.MessageId,
        ConversationId = msg.ConversationId,
        SenderUserId = msg.SenderUserId,
        Content = msg.Content,
        SentAtUtc = msg.SentAtUtc
    });
}
```

Validation steps:

- Authentication verification
- Content validation
- Membership validation

5 Database Design

5.1 Message Entity

```
public class Message
{
    public Guid Id { get; set; }

    public Guid ConversationId { get; set; }
    public Conversation? Conversation { get; set; }

    public Guid SenderUserId { get; set; }
    public User? SenderUser { get; set; }

    public DateTime SentAtUtc { get; set; }
    public string Content { get; set; } = string.Empty;
}
```

5.2 Fluent Configuration

```
modelBuilder.Entity<Message>()
    .HasOne(m => m.Conversation)
    .WithMany(c => c.Messages)
    .HasForeignKey(m => m.ConversationId)
    .OnDelete(DeleteBehavior.Cascade);

modelBuilder.Entity<Message>()
    .HasOne(m => m.SenderUser)
    .WithMany()
    .HasForeignKey(m => m.SenderUserId)
    .OnDelete(DeleteBehavior.Restrict);
```

5.3 Database Persistence

```
public async Task<MessageRecord> AddMessageAsync(
    Guid conversationId,
    Guid senderUserId,
    string content)
{
    var message = new Message
    {
        Id = Guid.NewGuid(),
        ConversationId = conversationId,
        SenderUserId = senderUserId,
        Content = content,
        SentAtUtc = DateTime.UtcNow
    };

    _db.Messages.Add(message);
    await _db.SaveChangesAsync();

    return ToRecord(message);
}
```

EF Core ensures transactional consistency and referential integrity.

6 Security Design

6.1 Conversation Membership Validation

```
if (!convo.ParticipantUserIds.Contains(senderUserId))
    throw new InvalidOperationException(
        "Sender is not a participant of this conversation.");
```

This prevents unauthorized message injection.

7 Migration from JSON to Database

The project initially used JSON-based file storage. Limitations included lack of foreign key constraints, full file rewrites and potential race conditions.

Replacing:

- IUserStore → DatabaseUserStore
- IChatStore → DatabaseChatStore

required no changes to controller logic due to interface abstraction.

8 Conclusion

Fusion demonstrates a structured and extensible client-server architecture built with modern software engineering principles. The project combines MVVM on the client side, layered API design on the server side, and relational persistence using PostgreSQL.

Security is enforced through JWT authentication, explicit authorization checks, and database-level referential integrity.

The architecture is scalable, maintainable and extensible, providing a solid foundation for future real-time features, fine-grained authorization and advanced database security mechanisms.